

## Official Transcript of Records

### ESA Academy Training Course:

*Earth Observation Satellite System Design Training Course 2023*

### Student personal data:

Complete name: Leo Bougault

Nationality: France

### Course data:

Location: **Training and Learning Facility, ESEC-Galaxia, Belgium**

Date: **6 November – 10 November 2023**

Total duration: **34.5 h**

| Lecture                                                                           | Duration   |
|-----------------------------------------------------------------------------------|------------|
| Welcome and Introduction                                                          | 45 min     |
| ESA, ESA Education & Early Career Opportunities at ESA                            | 45 min     |
| Introduction to Earth Observation (EO) & EO Satellite Systems                     | 1 h 30 min |
| From User Needs to System Requirements                                            | 1 h 30 min |
| Satellite Observation System Fundamentals in Earth Observation                    | 1 h 30 min |
| Use of EO Satellite Data in Science, Institutional Services and Commercial Market | 1 h        |
| Climate Change and its monitoring using Satellite Observations                    | 45 min     |
| Electromagnetic Wave Theory and Antennas                                          | 1 h 30 min |
| Radar Remote Sensing                                                              | 2 h        |
| Microwave Radiometry                                                              | 1 h 30 min |
| Basics of Space Optics                                                            | 1 h 30 min |
| Passive Optical Payloads                                                          | 1 h 30 min |
| Space Lidars                                                                      | 1 h        |
| Risk Management & Technology Development                                          | 1 h 30 min |
| Satellite System Design & Payload Accommodation                                   | 2 h 15 min |
| On-Ground AIV                                                                     | 1 h        |
| Ground Segment & Operations Concepts                                              | 1 h 30 min |
| Launch campaign                                                                   | 45 min     |
| In-Orbit Verification                                                             | 45 min     |
| Visit of ESEC-Redu                                                                | 1 h        |
| Group Project                                                                     | 5 h 45 min |
| Evaluation and Conclusions                                                        | 3 h 15 min |

## Evaluation result:

Grading system:

Grade: A

| Grade | Definition                                           | Grade | Definition                                           |
|-------|------------------------------------------------------|-------|------------------------------------------------------|
| A     | Outstanding performance with only minor errors       | D     | Fair but with significant shortcomings               |
| B     | Above the average standard but with some errors      | E     | Performance meets the minimum criteria               |
| C     | Generally sound work with a number of notable errors | F     | Fail – performance did not meet the minimum criteria |

Transinne, 13 November 2023



Hugo Marée  
Head of Education Office  
European Space Agency